🎭 Dual-Stream Deepfake Detection Engine: Master Technical Guide

📖 Complete Curriculum Index

This repository contains an exhaustive, zero-assumptions technical curriculum covering every physical, mathematical, architectural, and operational aspect of the Dual-Stream Deepfake Detection Engine.

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│                              CURRICULUM DIRECTORY                               │
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│ 📄 SECTION 1: Foundations & First Principles from Zero (Image Physics to Dual-Domain) │
│ 📄 SECTION 2: The Neural Network Architecture (ConvNeXt + SRM + Bayar + 2D FFT + Gate) │
│ 📄 SECTION 3: Face Preprocessing, Alignment & Zero-Leakage Graph Partitioning   │
│ 📄 SECTION 4: Training Engine & Optimization (DDP, SyncBatchNorm, Loss Masking, EMA) │
│ 📄 SECTION 5: Calibration, Temporal Video Aggregation & Inference Engine (L-BFGS-B) │
│ 📄 SECTION 6: Forensic Explainability & Interpretability Engine (Grad-CAM, SRM, FFT) │
│ 📄 SECTION 7: Quantitative Benchmarks, LOTO Generalization, Robustness & Latency │
│ 📄 SECTION 8: Interactive Web Application & Production Deployment (Streamlit)   │
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📑 Direct Section Links

1. Section 1: Foundations & First Principles
2. Hardware image capture, CMOS sensors, PRNU noise physics.
3. The 5 generative manipulation technologies (*Deepfakes, Face2Face, FaceSwap, NeuralTextures, Celeb-DF v2*).
4. Forensic flaws (PRNU disruption, checkerboard grids, boundary resolution mismatches).
5. 2D Discrete Fourier Transform mathematics, Euler's formula, and steganographic principles.
6. Section 2: Neural Network Architecture
7. Spatial stream with `ConvNeXt-Small` and pre-classifier `LayerNorm2d(768)`.
8. Frequency stream with 9-channel `SRMConv2d`, 1-channel `BayarConv2d`, and 20-channel `RealFFT2DModule`.
9. Symmetric Gated Residual Fusion ($g \in [0,1]^{512}$, $\mathbf{f}_{\text{fused}} \in \mathbb{R}^{1024}$).
10. 5D sequence chunked processing (`forward_sequence`).
11. Section 3: Face Preprocessing & Alignment
12. Multi-engine face detection cascade (OpenCV YuNet $\rightarrow$ MTCNN $\rightarrow$ Haar Cascade $\rightarrow$ Center Crop).

13. $1.50\times$ bounding box scaling and 5-point LMEDS affine landmark alignment.
14. 2D Cosine window edge tapering eliminating FFT boundary cross-spikes.
15. Zero-leakage graph connected-component partitioning (`networkx.Graph`).
16. Section 4: Training Engine & Optimization
17. Multi-GPU Distributed Data Parallel (DDP) with Hugging Face `Accelerate` and `SyncBatchNorm` .
18. Per-sample loss masking (`valid_flags`) and dynamic `pos_weight` calculation.
19. 4-way differential parameter grouping (`lr_backbone = 1e-4` , `lr_head = 1e-3`).
20. Exponential Moving Average (EMA) shadow weight updates ($\beta = 0.999$).
21. Section 5: Calibration & Temporal Aggregation
22. SciPy L-BFGS-B log-temperature scaling ($T^* = 1.4788$, ECE reduction: $0.0122 \to 0.0093$).
23. Operational confidence score normalization ($[50.0\%, 100.0\%]$ anchored at threshold $\theta = 0.01$).
24. 4 temporal aggregation algorithms: Softmax-Weighted ($\tau = 0.10$), Top-k, EMA, and Mean.
25. Sequential video decoding with Test-Time Augmentation (TTA).
26. Section 6: Forensic Explainability & Interpretability
27. ConvNeXt Stage 4 Grad-CAM hooks and gradient-weighted activation mapping.
28. 9-channel SRM noise residual maps rendered with OpenCV Viridis.
29. Centered 2D Real FFT log-magnitude spectrums rendered with OpenCV Magma.
30. Publication-ready 4-panel diagnostic visualizer (`visualize_attention_maps.py`).
31. Section 7: Benchmarks, Robustness & Latency
32. Held-out test set performance: **ROC AUC** `0.9988` , **F1** `0.9830` , **Recall** `99.79%` .
33. Leave-One-Target-Out (LOTO) cross-generator zero-shot generalization ($0.9662 - 0.9783$ AUC).
34. Degradation stress tests (JPEG, Blur, Noise, Downscaling) and failure mode analysis.
35. Hardware latency benchmarks on NVIDIA Tesla T4 GPU (18.62 ms) and Intel Xeon CPU (4.77 ms).
36. Section 8: Web Application & Production Deployment
37. Full-stack Streamlit web app with singleton model caching and CUDA warm-up.
38. Dark glassmorphism UI/UX styling and mobile responsive breakpoints.
39. Interactive temporal anomaly sequence timeline plotting in Matplotlib.
40. Production Docker containerization and Hugging Face Spaces cloud serving.

🏛️ Master Architecture Diagram

```
[ Input Video / Image ] ──▶ [ OpenCV YuNet Alignment ] ──▶ [ 512x512 Face Crops (1.50x Scale
                                                                  │
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                         ▼
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```
             [ Spatial Stream (ConvNeXt-Small) ]                                [ Frequency Stream
             • 256x256 RGB Image Input                                          • 3 Fixed SRM Filte
             • LayerNorm2d Spatial Normalization                                • 20-Channel 2D FF
             • 512-d Feature Embedding f_s                                      • 512-d Frequency 
                            |
                            └────────────────────────────────┬──────────────────────────────────
                                                             ▼
                                        [ Symmetric Gated Residual Fusion ]
                                        • Gating Vector g = Sigmoid(Linear(f_s || f_f)) in 
                                        • Fused Feature f_fused = [f_s * (1 - g) || f_f * g
                                                             |
                                                             ▼
                                            [ Binary Classifier Head ]
                                            • Linear(1024, 256) + Dropout(0.3) + Linear(256
                                            • Raw Logit Output z
                                                             |
                            ┌────────────────────────────────┴───────────────────────────────┐
                            ▼                                                                ▼
         [ SciPy L-BFGS-B Temperature Calibration ]                                [ 4-Panel Interpretability
         • Calibrated Logit: z / T* (T* = 1.4788)                                 • (a) RGB Input Face Crop
         • Probability: p = Sigmoid(z / T*)                                         • (b) SRM Noise Residual M
         • Expected Calibration Error: 0.0122 → 0.0093                              • (c) 2D FFT Magnitude Spe
                                                                                    • (d) ConvNeXt Grad-CAM At
```

📐 Master Mathematical Formula Sheet

1. 2D Discrete Fourier Transform (Spectral Decomposition)

$$F(u,v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} f(x,y) \cdot e^{-j2\pi\left(\frac{ux}{H} + \frac{vy}{W}\right)}$$

$$\mathcal{M}(u,v) = \ln\Big(\big|\text{fftshift}(F(u,v))\big| + 1\Big), \quad \Phi(u,v) = \frac{\text{angle}(\text{fftshift}(F(u,v)))}{\pi}$$

2. Bayar-Stamm Adaptive Constrained Convolution

$$\mathbf{W}*{|text{constrained}}(i, j) = |begin{cases} -1.0, & |text{if } (i, j) = (0, 0) | |frac{|mathbf{W}(i, j)} {|sum*{(u,v) \neq (0,0)} \mathbf{W}(u, v)}, & \text{if } (i, j) \neq (0, 0) \end{cases}$$

3. Symmetric Gated Residual Fusion

$$g = \sigma\Big(\mathbf{W}*g [|mathbf{f}*{\text{spatial}} \;\|\; \mathbf{f}*{|text{freq}}] + |mathbf{b}_g |Big) |in [0, 1]^{512}*

|mathbf{f}{\text{fused}} = \Big[\mathbf{f}*{|text{spatial}} |odot (1 - g) |;\|; |mathbf{f}*{\text{freq}} \odot g \Big] \in \mathbb{R}^{1024}$$

4. Per-Sample Masked Binary Cross-Entropy Loss

$$\mathcal{L} = \frac{\sum_{i=1}^B \mathcal{L}*{|text{BCE}}(z_i, y_i) |cdot v_i}{|max|left(1, |sum*{i=1}^B v_i\right)}, \quad v_i \in {0.0, 1.0}$$

5. Temperature Scaling Calibration ($T^* = 1.4788$)

$$p_{\text{calibrated}} = \sigma\left(\frac{z}{T^*} |right) = |frac{1}{1 + e^{-z / T^*}}$$

6. Softmax-Weighted Video Temporal Aggregation ($\tau = 0.10$)

$$S_{\text{video}} = \sum_{k=1}^{K} w_k \cdot p_k, \quad \text{where } w_k = \frac{\exp\left(\frac{p_k}{\tau}\right)}{\sum_{j=1}^{K} \exp\left(\frac{p_j}{\tau}\right)}$$

⚡ Quickstart Commands

```
# 1. Clone repository and install dependencies
git clone https://github.com/yyouretoast/deepfake-detection.git
cd deepfake-detection
pip install -r requirements.txt

# 2. Run the full automated test suite (66/66 passing tests)
pytest tests/ -v

# 3. Launch the interactive Streamlit Web Application
streamlit run app.py

# 4. Run Multi-GPU DDP Training
accelerate launch --mixed_precision fp16 --num_processes 2 --multi_gpu scripts/train_dual_st

# 5. Generate Publication-Ready Benchmark Figures
python scripts/generate_benchmark_plots.py
```
